РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №10

дисциплина: Администрирование сетевых подсистем

Студент: Талебу Тенке

Студ. билет № 1032224534

Группа: НФИбд-02-23

МОСКВА

2025г.

Цель работы:

Целью данной работы является приобретение практических навыков по конфигурированию SMTP-сервера в части настройки аутентификации.

Выполнение работы:

Open Save

10-master.conf
/etc/dovecot/conf.d

```
    #ssl = yes
  }
}

service submission-login {
  inet_listener submission {
    #port = 587
  }
}

service lmtp {
        unix_listener /var/spool/postfix/private/dovecot-lmtp {
        group = postfix
        user = postfix
        mode = 0600
  }
}

service imap {
  # Most of the memory goes to mmap()ing files. You may need to increase this
  # limit if you have huge mailboxes.
  #vsz_limit = $default_vsz_limit

  # Max. number of IMAP processes (connections)
  #process_limit = 1024
}

service pop3 {
  # Max. number of POP3 processes (connections)
  #process_limit = 1024
}

service submission {
  # Max. number of SMTP Submission processes (connections)
  #process_limit = 1024
}

service auth {
```

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10-auth.conf
/etc/dovecot/conf.d

```
# value contains series of from -> to characters. For example "#@/@" means
# that '#' and '/' characters are translated to '@'.
#auth_username_translation =

# Username formatting before it's looked up from databases. You can use
# the standard variables here, eg. %Lu would lowercase the username, %n would
# drop away the domain if it was given, or "%n-AT-%d" would change the '@' into
# "-AT-". This translation is done after auth_username_translation changes.
#auth_username_format = %Ln

# If you want to allow master users to log in by specifying the master
# username within the normal username string (ie. not using SASL mechanism's
# support for it), you can specify the separator character here. The format
# is then <username><separator><master username>. UW-IMAP uses "*" as the
# separator, so that could be a good choice.
#auth_master_user_separator =

# Username to use for users logging in with ANONYMOUS SASL mechanism
#auth_anonymous_username = anonymous

# Maximum number of dovecot-auth worker processes. They're used to execute
# blocking passdb and userdb queries (eg. MySQL and PAM). They're
# automatically created and destroyed as needed.
#auth_worker_max_count = 30

# Host name to use in GSSAPI principal names. The default is to use the
# name returned by gethostname(). Use "$ALL" (with quotes) to allow all keytab
# entries.
#auth_gssapi_hostname =

# Kerberos keytab to use for the GSSAPI mechanism. Will use the system
# default (usually /etc/krb5.keytab) if not specified. You may need to change
# the auth service to run as root to be able to read this file.
#auth_krb5_keytab =
```

```
root@server:~
root@server:/etc/dovecot/conf.d    root@server:~
[vagrant@server ~]$ sudo -i
[root@server ~]# MAIL=~/Maildir/ mai
bash: mai: command not found...
[root@server ~]# MAIL=~/Maildir/ mail
s-nail: No mail for root at /root/Maildir/
s-nail: /root/Maildir/: No such entry, file or directory
[root@server ~]#
```

```
** (gedit:3411): WARNING **: 11:19:25.774: Set document metadata failed: Setting attribute m
etadata::gedit-spell-language not supported

** (gedit:3411): WARNING **: 11:19:25.779: Set document metadata failed: Setting attribute m
etadata::gedit-encoding not supported

** (gedit:3411): WARNING **: 11:19:38.413: Set document metadata failed: Setting attribute m
etadata::gedit-spell-language not supported

** (gedit:3411): WARNING **: 11:19:38.415: Set document metadata failed: Setting attribute m
etadata::gedit-encoding not supported

** (gedit:3411): WARNING **: 11:19:39.271: Set document metadata failed: Setting attribute m
etadata::gedit-position not supported

(gedit:3411): dconf-WARNING **: 11:19:39.291: failed to commit changes to dconf: Failed to e
xecute child process “dbus-launch” (No such file or directory)
[root@server conf.d]# systemctl restart postfix
[root@server conf.d]# systemctl restart dovecot
[root@server conf.d]# echo .| mail -s "LMTP test" user@user.net
[root@server conf.d]#
```

Open Save

10-master.conf
/etc/dovecot/conf.d

```
  #process_limit = 1024
}

service submission {
  # Max. number of SMTP Submission processes (connections)
  #process_limit = 1024
}

service auth {
        unix_listener /var/spool/postfix/private/auth {
        group = postfix
        user = postfix
        mode = 0660
        }
        unix_listener auth-userdb {
        mode = 0600
        user = dovecot
 }
}

  # Postfix smtp-auth
  #unix_listener /var/spool/postfix/private/auth {
  #  mode = 0666
  #}

  # Auth process is run as this user.
  #user = $default_internal_user
}

service auth-worker {
  # Auth worker process is run as root by default, so that it can access
  # /etc/shadow. If this isn't necessary, the user should be changed to
  # $default_internal_user.
  #user = root
}

service dict {
```

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```
root@server:/etc/dovecot/conf.d
root@server:/etc/dovecot/conf.d        root@server:~
** (gedit:3650): WARNING **: 11:30:26.889: Set document metadata failed: Setting attribut
etadata::gedit-encoding not supported

** (gedit:3650): WARNING **: 11:30:29.288: Set document metadata failed: Setting attribut
etadata::gedit-spell-language not supported

** (gedit:3650): WARNING **: 11:30:29.290: Set document metadata failed: Setting attribut
etadata::gedit-encoding not supported

** (gedit:3650): WARNING **: 11:30:30.057: Set document metadata failed: Setting attribut
etadata::gedit-spell-language not supported

** (gedit:3650): WARNING **: 11:30:30.061: Set document metadata failed: Setting attribut
etadata::gedit-encoding not supported

** (gedit:3650): WARNING **: 11:30:30.867: Set document metadata failed: Setting attribut
etadata::gedit-position not supported

(gedit:3650): dconf-WARNING **: 11:30:30.941: failed to commit changes to dconf: Failed t
xecute child process “dbus-launch” (No such file or directory)
[root@server conf.d]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server conf.d]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions =
reject_unknown_recipient_domain, permit_mynetworks,
reject_non_fqdn_recipient, reject_unauth_destination,
reject_unverified_recipient, permit'
postconf: fatal: -e, -X, or -# accepts no multi-line input
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipien
omain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_un
ified_recipient, permit'
[root@server conf.d]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server conf.d]#
```

Open ▾ | master.cf /etc/postfix | Save

```
1 #
2 # Postfix master process configuration file.  For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # ==========================================================================
9 # service type  private unpriv  chroot  wakeup  maxproc command + args
10 #               (yes)   (yes)   (no)    (never) (100)
11 # ==========================================================================
12 smtp      inet  n       -       n       -       -       smtpd
13 #smtp      inet  n       -       n       -       1       postscreen
14 #smtpd     pass  -       -       n       -       -       smtpd
15 #dnsblog   unix  -       -       n       -       0       dnsblog
16 #tlsproxy  unix  -       -       n       -       0       tlsproxy
17 #submission inet n       -       n       -       -       smtpd
18 #  -o syslog_name=postfix/submission
19 #  -o smtpd_tls_security_level=encrypt
20 #  -o smtpd_sasl_auth_enable=yes
21 #  -o smtpd_tls_auth_only=yes
22 #  -o smtpd_reject_unlisted_recipient=no
23 #  -o smtpd_client_restrictions=$mua_client_restrictions
24 #  -o smtpd_helo_restrictions=$mua_helo_restrictions
25 #  -o smtpd_sender_restrictions=$mua_sender_restrictions
26 #  -o smtpd_recipient_restrictions=
27 #  -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
28 #  -o milter_macro_daemon_name=ORIGINATING
29 #smtps     inet  n       -       n       -       -       smtpd
30 #  -o syslog_name=postfix/smtps
31 #  -o smtpd_tls_wrappermode=yes
32 #  -o smtpd_sasl_auth_enable=yes
33 #  -o smtpd_reject_unlisted_recipient=no
34 #  -o smtpd_client_restrictions=$mua_client_restrictions
35 #  -o smtpd_helo_restrictions=$mua_helo_restrictions
36 #  -o smtpd_sender_restrictions=$mua_sender_restrictions
37 #  -o smtpd_recipient_restrictions=
```

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Open ▾ *master.cf /etc/postfix Save

```
1 #
2 # Postfix master process configuration file.  For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # ==========================================================================
9 # service type  private unpriv  chroot  wakeup  maxproc command + args
10 #               (yes)   (yes)   (no)    (never) (100)
11 # ==========================================================================
12 smtp      inet  n       -       n       -       -       smtpd inet n - n - - smtpd o
   smtpd_sasl_auth_enable=yes
13 -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknow ]
14 n_recipient_domain,permit_sasl_authenticated,reject
15 #smtp      inet  n       -       n       -       1       postscreen
16 #smtpd     pass  -       -       n       -       -       smtpd
17 #dnsblog   unix  -       -       n       -       0       dnsblog
18 #tlsproxy  unix  -       -       n       -       0       tlsproxy
19 #submission inet n       -       n       -       -       smtpd
20 #  -o syslog_name=postfix/submission
21 #  -o smtpd_tls_security_level=encrypt
22 #  -o smtpd_sasl_auth_enable=yes
23 #  -o smtpd_tls_auth_only=yes
24 #  -o smtpd_reject_unlisted_recipient=no
25 #  -o smtpd_client_restrictions=$mua_client_restrictions
26 #  -o smtpd_helo_restrictions=$mua_helo_restrictions
27 #  -o smtpd_sender_restrictions=$mua_sender_restrictions
28 #  -o smtpd_recipient_restrictions=
29 #  -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
30 #  -o milter_macro_daemon_name=ORIGINATING
31 #smtps     inet  n       -       n       -       -       smtpd
32 #  -o syslog_name=postfix/smtps
33 #  -o smtpd_tls_wrappermode=yes
34 #  -o smtpd_sasl_auth_enable=yes
35 #  -o smtpd_reject_unlisted_recipient=no
36 #  -o smtpd_client_restrictions=$mua_client_restrictions
```

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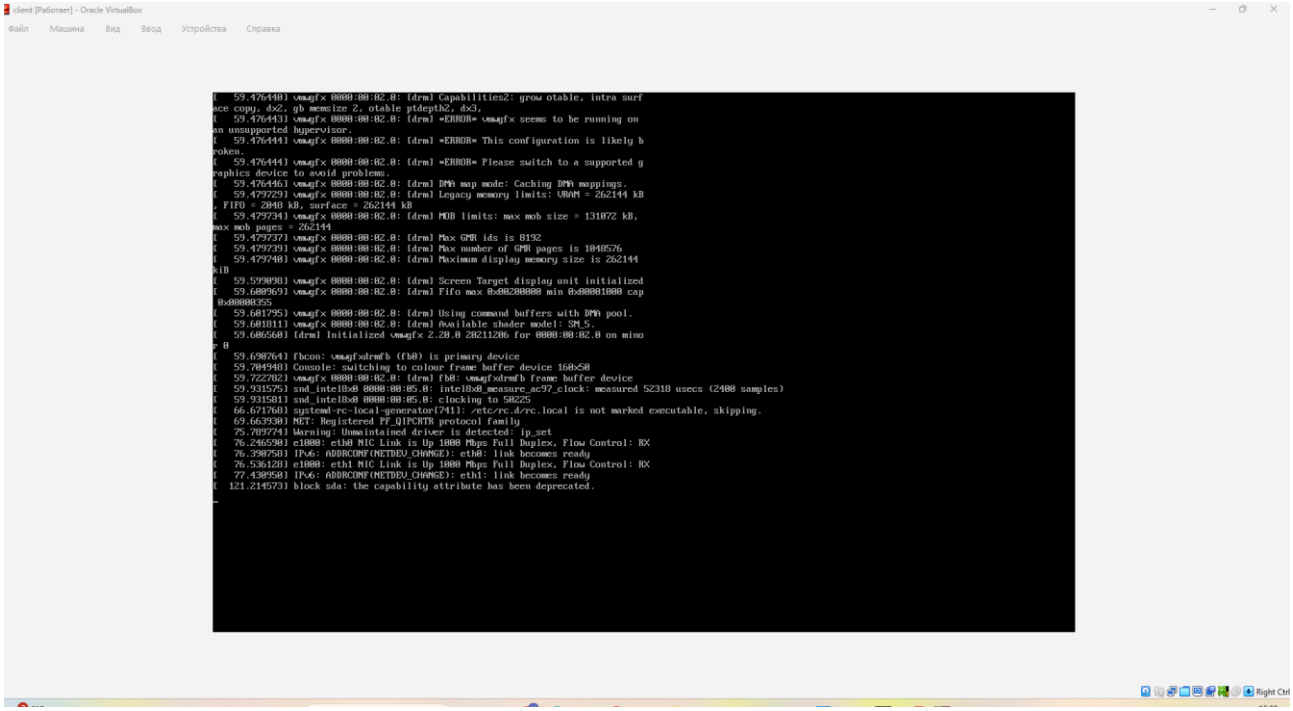
```
FarIta.lng
FarLit.lng
FarPol.hlf
FarPol.lng
FarPor.lng
FarRus.hlf
FarRus.lng
FarSky.hlf
..                    Up   20.02.26 09:45   ..                    Up   25.02.26 13:28
Bytes: 6,30 K, files: 2, folders: 2         Bytes: 19,6 M, files: 34, folders: 5
C:\work\frank taleubou\vagrant>  vagrant dele
1Help  2UserMn 3View  4Edit  5Copy  6RenMov 7MkFold 8Delete 9ConfMn 10Quit  11Plugin
```



client [Работает] - Oracle VirtualBox

Файл Машина Вид Ввод Устройства Справка

Activities Terminal Mar 11 12:10 en

root@client:~

```
[vagrant@client ~]$ sudo -i
[root@client ~]# dnf -y install telnet
Extra Packages for Enterprise Linux 9 - x86_64                27 kB/s |  28 kB     00:01
Extra Packages for Enterprise Linux 9 - x86_64               204 kB/s |  20 MB     01:42
Extra Packages for Enterprise Linux 9 openh264 (From Cisco) - 1.6 kB/s | 993  B     00:00
Rocky Linux 9 - BaseOS                                       5.8 kB/s | 4.3 kB     00:00
Rocky Linux 9 - BaseOS                                       4.9 MB/s |  16 MB     00:03
Rocky Linux 9 - AppStream                                     11 kB/s | 4.8 kB     00:00
Rocky Linux 9 - AppStream                                    4.7 MB/s |  16 MB     00:03
Rocky Linux 9 - Extras                                       561  B/s | 3.1 kB     00:05
Rocky Linux 9 - Extras                                        21 kB/s |  17 kB     00:00
Dependencies resolved.
================================================================================================
 Package            Architecture       Version                    Repository            Size
================================================================================================
Installing:
 telnet             x86_64             1:0.17-85.el9              appstream             63 k

Transaction Summary
================================================================================================
Install  1 Package

Total download size: 63 k
Installed size: 121 k
Downloading Packages:
telnet-0.17-85.el9.x86_64.rpm                                 95 kB/s |  63 kB     00:00
------------------------------------------------------------------------------------------------
Total                                                         49 kB/s |  63 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
```

Right Ctrl

client [Работает] - Oracle VirtualBox

Файл Машина Вид Ввод Устройства Справка

Activities Terminal Mar 11 12:16 en

root@client:~

```
--------------------------------------------------------------------------------
Total                                                   49 kB/s |  63 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                          1/1
  Installing       : telnet-1:0.17-85.el9.x86_64                              1/1
  Running scriptlet: telnet-1:0.17-85.el9.x86_64                              1/1
  Verifying        : telnet-1:0.17-85.el9.x86_64                              1/1

Installed:
  telnet-1:0.17-85.el9.x86_64

Complete!
[root@client ~]# print 'vagrant\x00vagrant\x00123456'
bash: print: command not found...
[root@client ~]# telnet server.user.net 25
telnet: server.user.net: Name or service not known
server.user.net: Unknown host
[root@client ~]# telnet server.vagrant.net 25
Trying 13.248.169.48...
telnet: connect to address 13.248.169.48: Connection refused
Trying 76.223.54.146...
telnet: connect to address 76.223.54.146: Connection refused
[root@client ~]# EHLO test
bash: EHLO: command not found...
[root@client ~]# AUTH PLAIN <>
-bash: syntax error near unexpected token `newline'
[root@client ~]# AUTH PLAIN dXNlcgB1c2VyADEyMzQ1Ng==
bash: AUTH: command not found...
[root@client ~]#
```

Right Ctrl

master.cf
/etc/postfix

```
#
# ==========================================================================
#
# Old example of delivery via Cyrus.
#
#old-cyrus unix  -       n       n       -       -       pipe
#  flags=R user=cyrus argv=/usr/lib/cyrus-imapd/deliver -e -m ${extension} ${user}
#
# ==========================================================================
#
# See the Postfix UUCP_README file for configuration details.
#
#uucp      unix  -       n       n       -       -       pipe
#  flags=Fqhu user=uucp argv=uux -r -n -z -a$sender - $nexthop!rmail ($recipient)
#
# ==========================================================================
#
# Other external delivery methods.
#
#ifmail    unix  -       n       n       -       -       pipe
#  flags=F user=ftn argv=/usr/lib/ifmail/ifmail -r $nexthop ($recipient)
#
#bsmtp     unix  -       n       n       -       -       pipe
#  flags=Fq. user=bsmtp argv=/usr/local/sbin/bsmtp -f $sender $nexthop $recipient
#
#scalemail-backend unix -       n       n       -       2       pipe
#  flags=R user=scalemail argv=/usr/lib/scalemail/bin/scalemail-store
#  ${nexthop} ${user} ${extension}
#
#mailman   unix  -       n       n       -       -       pipe
#  flags=FRX user=list argv=/usr/lib/mailman/bin/postfix-to-mailman.py
#  ${nexthop} ${user}
submission inet n - n - - smtpd
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
-o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknow
n_recipient_domain,permit_sasl_authenticated,reject
```

Plain Text Tab Width: 8 Ln 134, Col 32 INS

root@server:~

root@server:/etc/dovecot/conf.d | root@server:~

```
s-nail: No mail for root at /root/Maildir/
s-nail: /root/Maildir/: No such entry, file or directory
[root@server ~]# dnf -y install telnet
Rocky Linux 9 - BaseOS                                  5.9 kB/s | 4.3 kB     00:00
Rocky Linux 9 - AppStream                               6.0 kB/s | 4.8 kB     00:00
Rocky Linux 9 - Extras                                  3.0 kB/s | 3.1 kB     00:01
Package telnet-1:0.17-85.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server ~]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs
[root@server ~]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private
[root@server ~]# postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'
[root@server ~]# postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'
[root@server ~]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtp
d_scache'
[root@server ~]# postconf -e 'smtpd_tls_security_level = may
> ^C
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'
[root@server ~]# postconf -e 'smtp_tls_security_level = may'
```

```
root@server:~
root@server:/etc/dovecot/conf.d        root@server:~
ix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd ne
bula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovir
t-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql
privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio pupp
etmaster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-maste
r samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmp
tls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming s
vdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle t
ftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpinator wbem-ht
tp wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp w
sman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server ze
rotier
[root@server ~]# firewall-cmd --add-service=smtp-submission
success
[root@server ~]# firewall-cmd --add-service=smtp-submission --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# systemctl restart postfix
Job for postfix.service failed because the control process exited with error code.
See "systemctl status postfix.service" and "journalctl -xeu postfix.service" for details.
[root@server ~]# openssl s_client -starttls smtp -crlf -connect server.vagrant.net:587
801B44331D7F0000:error:8000006F:system library:BIO_connect:Connection refused:crypto/bio/bio
_sock2.c:114:calling connect()
801B44331D7F0000:error:10000067:BIO routines:BIO_connect:connect error:crypto/bio/bio_sock2.
c:116:
801B44331D7F0000:error:8000006F:system library:BIO_connect:Connection refused:crypto/bio/bio
_sock2.c:114:calling connect()
801B44331D7F0000:error:10000067:BIO routines:BIO_connect:connect error:crypto/bio/bio_sock2.
c:116:
connect:errno=111
[root@server ~]#
```

```
mail.sh
/vagrant/provision/server
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost,
$mydomain'↪
#postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
echo "Configure postfix for dovecot"
postconf -e 'home_mailbox = Maildir/'
echo "Configure postfix for auth"
postconf -e 'smtpd_sasl_type = dovecot'
postconf -e 'smtpd_sasl_path = private/auth'
Королькова А. В., Кулябов Д. С. Администрирование сетевых подсистем 81
postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain,
permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination,
reject_unverified_recipient, permit'
↪
↪
postconf -e 'mynetworks = 127.0.0.0/8'
echo "Configure postfix for SMTP over TLS"
cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs
cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private
postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'
postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'
postconf -e 'smtpd_tls_session_cache_database =
btree:/var/lib/postfix/smtpd_scache'↪
postconf -e 'smtpd_tls_security_level = may'
postconf -e 'smtp_tls_security_level = may'
postfix set-permissions
restorecon -vR /etc
systemctl stop postfix
systemctl start postfix
systemctl restart dovecot
sh  Tab Width: 8  Ln 42, Col 43  INS
```

Ответы на контрольные вопросы:

1. Приведите пример задания формата аутентификации пользователя в Dovecot в форме логина с указанием домена. –

   Допустим, у нас есть почтовый ящик с адресом user@example.com. В конфигурационном файле Dovecot (/etc/dovecot/conf.d/10-auth.conf), мы можем указать формат аутентификации следующим образом:

   auth_username_format = %Lu

В этом примере %Lu означает, что аутентификация будет проходить в формате "user" без учета регистра букв. Если вам нужно учитывать домен, вы можете использовать %n:

auth_username_format = %Ln

Таким образом, при вводе логина "user@example.com" пользователь будет аутентифицироваться с именем пользователя "user" и доменом "example.com".

2. Какие функции выполняет почтовый Relay-сервер? –

Пересылка почты: Relay-сервер принимает почтовые сообщения от клиентов и пересылает их к адресатам. Это особенно полезно, если у вас нет прямого доступа к серверу назначения или если вы хотите централизованно управлять отправкой почты.

Маршрутизация почты: Relay-сервер может определять наилучший маршрут для доставки почты на основе определенных правил и политик.

Блокировка спама: Некоторые Relay-серверы выполняют функции фильтрации спама, блокируя нежелательные сообщения до их отправки на сервер назначения.

3. Какие угрозы безопасности могут возникнуть в случае настройки почтового сервера как Relay-сервера? –

Открытый Relay: Если сервер настроен как открытый Relay, это может привести к злоупотреблению. Злоумышленники могут использовать сервер для отправки спама, что может повлечь за собой блокировку IP-адреса сервера или другие санкции.

Спуфинг: Атаки, связанные с подделкой отправителя (спуфинг), могут быть использованы для маскировки настоящего источника почты. Это может быть проблемой, если сервер Relay доверяет внешним источникам без должной аутентификации.

Отказ в обслуживании (DoS): Атаки типа DoS могут быть направлены на Relay-сервер, перегружая его запросами на пересылку почты и создавая неприемлемую загрузку.